Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*			Between 6:00 and 10:00 the wind speed is [just] above 3.5 m/s [or cut-in speed](1) so the power output will be low (1) After 16:00, power not produced until 19:00 (1) by 20:00 speed rises to about 4.2 m/s which still gives low power output (1) NB numerical information needed for marks 1, 3 and 4. Only award 4 marks if 'claim not supported' stated		1 1	1 1	4		
	(b)	(i)		Substitution into transformer equation e.g. $\frac{V_2}{16} = \frac{4800}{400}$ (1) or $V_2 = \frac{4800}{400} \times 16$ V_2 = 192 [kV] (1) c.a.o.		2		2	2	
		(ii)		Manipulation: $I = \frac{P}{V}$ (1) Substitution: $I = \frac{24[\times 10^6]}{192[\times 10^3]\textbf{ecf}}$ (1) Current = 125 [A] (1) NB. 1.25×10^n ($n \neq 2$) $\rightarrow$ 2 marks NB. $I = \frac{V}{R} = \frac{24[\times 10^6]}{192[\times 10^3]}$ = 125 $\rightarrow$ 0 marks		3		3	3	